Phys 153: Fundamentals of Physics III

Unit #4 – Lorentz Transformation

Aaron Wirthwein

Spring 2027

Standard Orientation (Conventions)

Time to roll up our sleeves. Before we get started, we adopt a few conventions that make life easier:

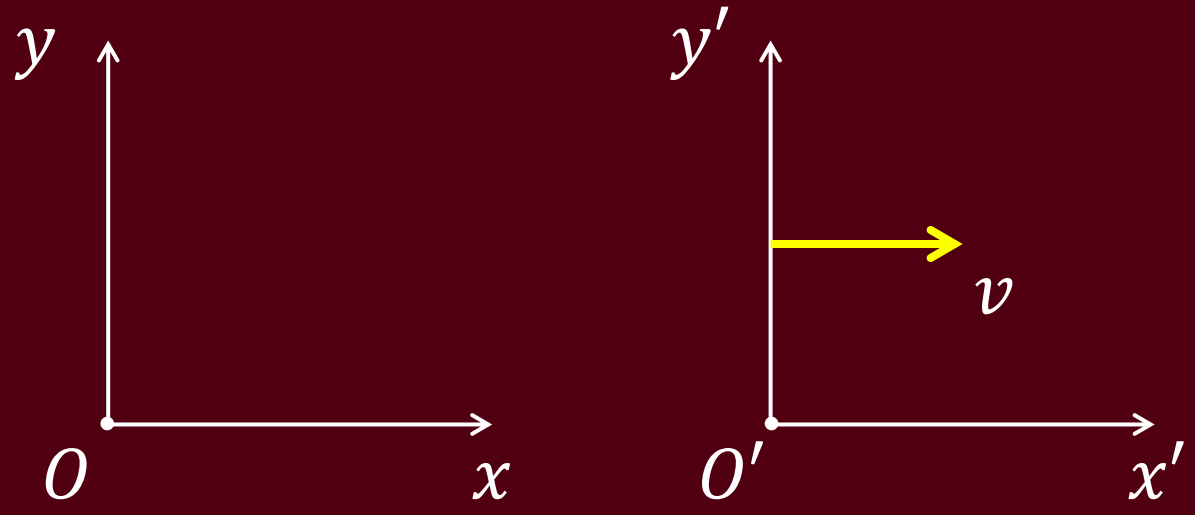
Home Frame: inertial reference frame S with origin O and spacetime coordinates (x, y, z, ct)

Other Frame: inertial frame S' with origin O' and spacetime coordinates (x', y', z', ct')

- The Other Frame moves with constant relative speed v along the x axis of the Home Frame.
- The spatial axes are aligned in both frames, e.g., the x' axis in S' points the same way as the x axis in S.
- The origins coincide at $t = t' = 0$; frames are synchronized

We will call this the "standard orientation."

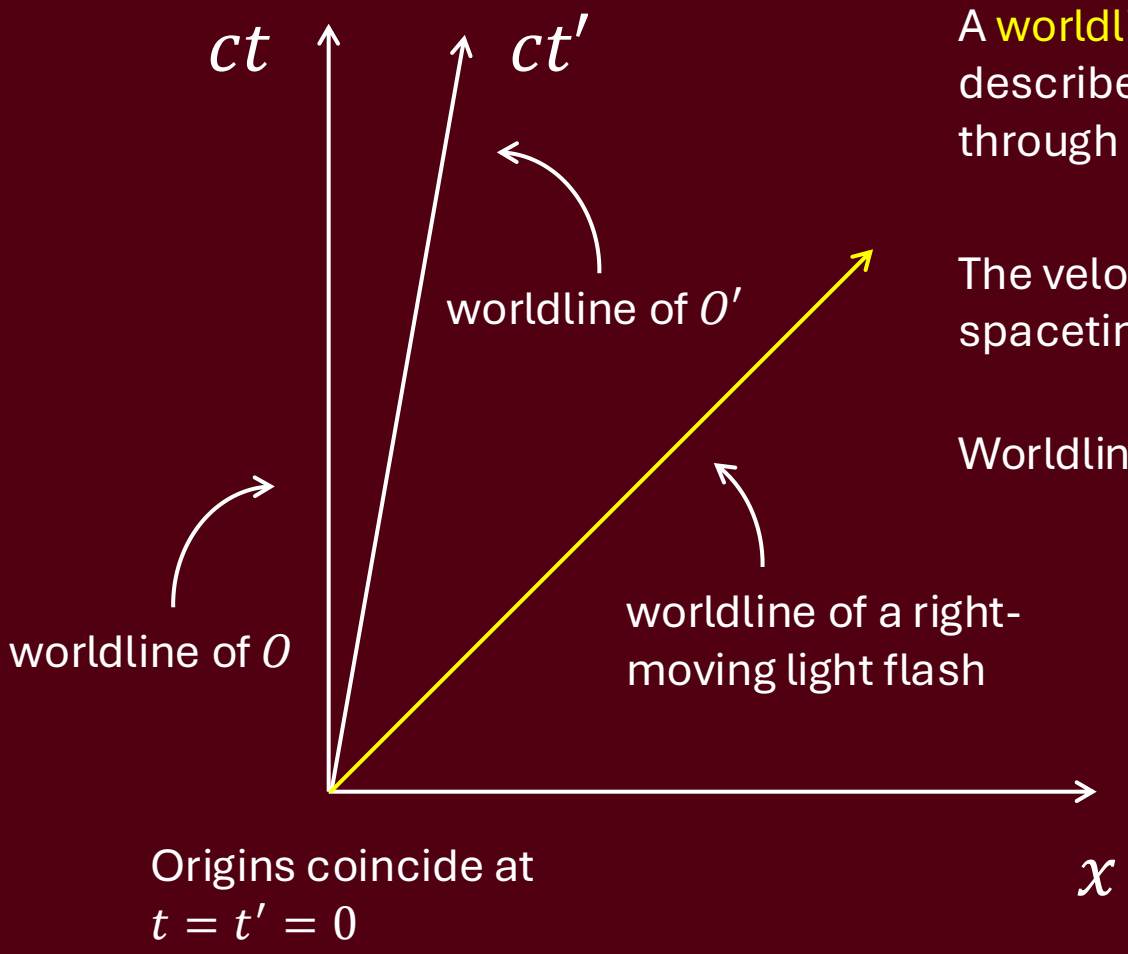
Standard Orientation



Home Frame **Other Frame**

The Other Frame moves at constant speed v relative to the Home Frame

Spacetime Diagram

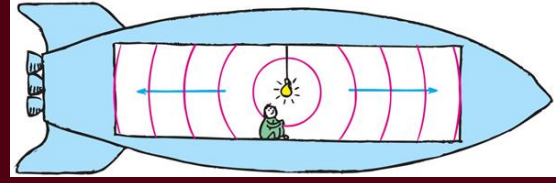


A worldline is a series of events that describes the trajectory of a particle through spacetime.

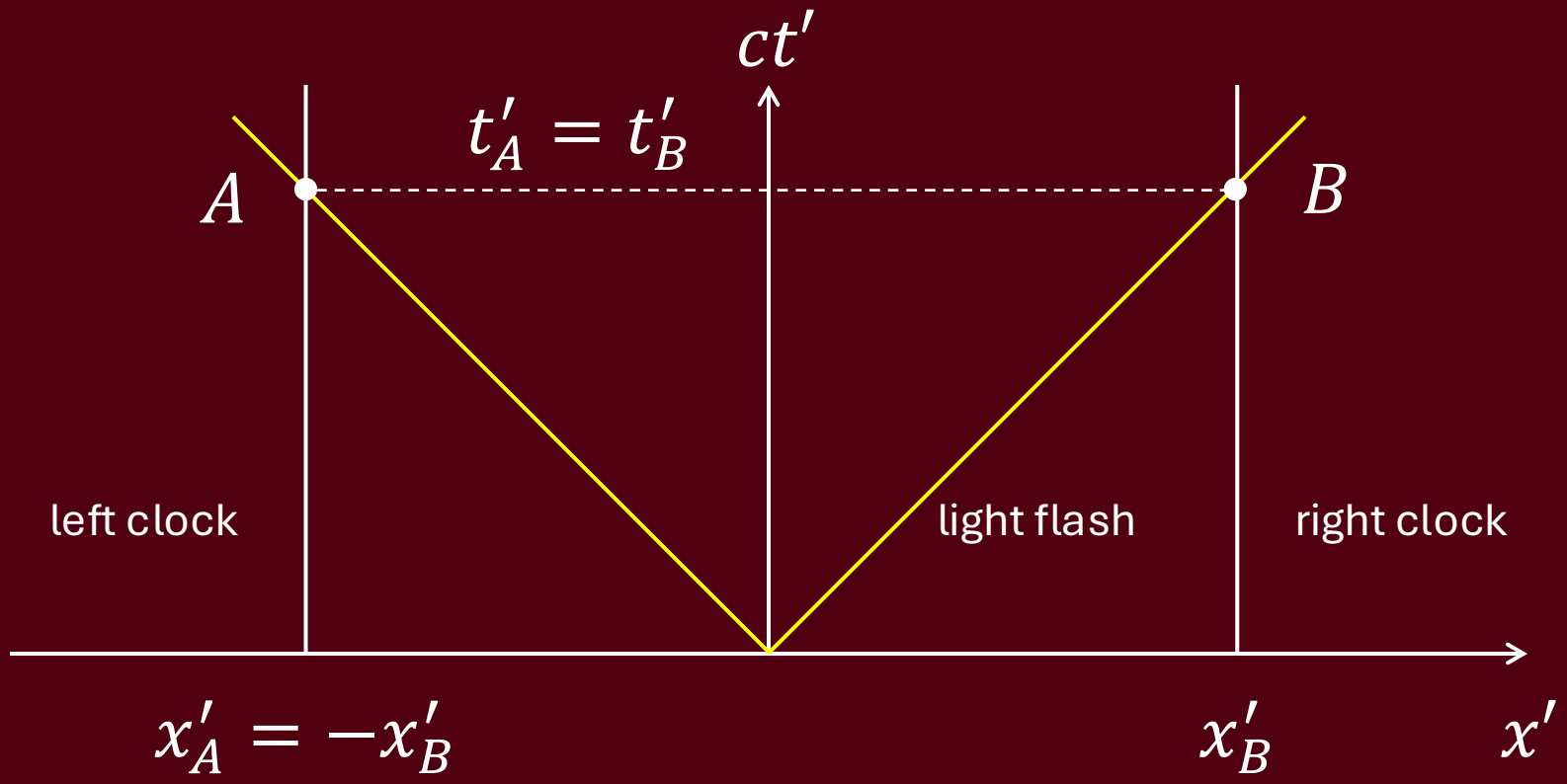
The velocity of a particle is 1/slope on a spacetime diagram, not slope

Worldline of light flash has slope ± 1

Relativity of Simultaneity

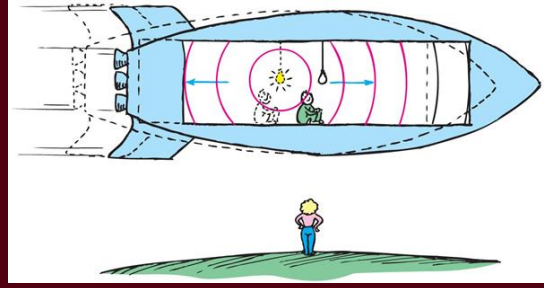


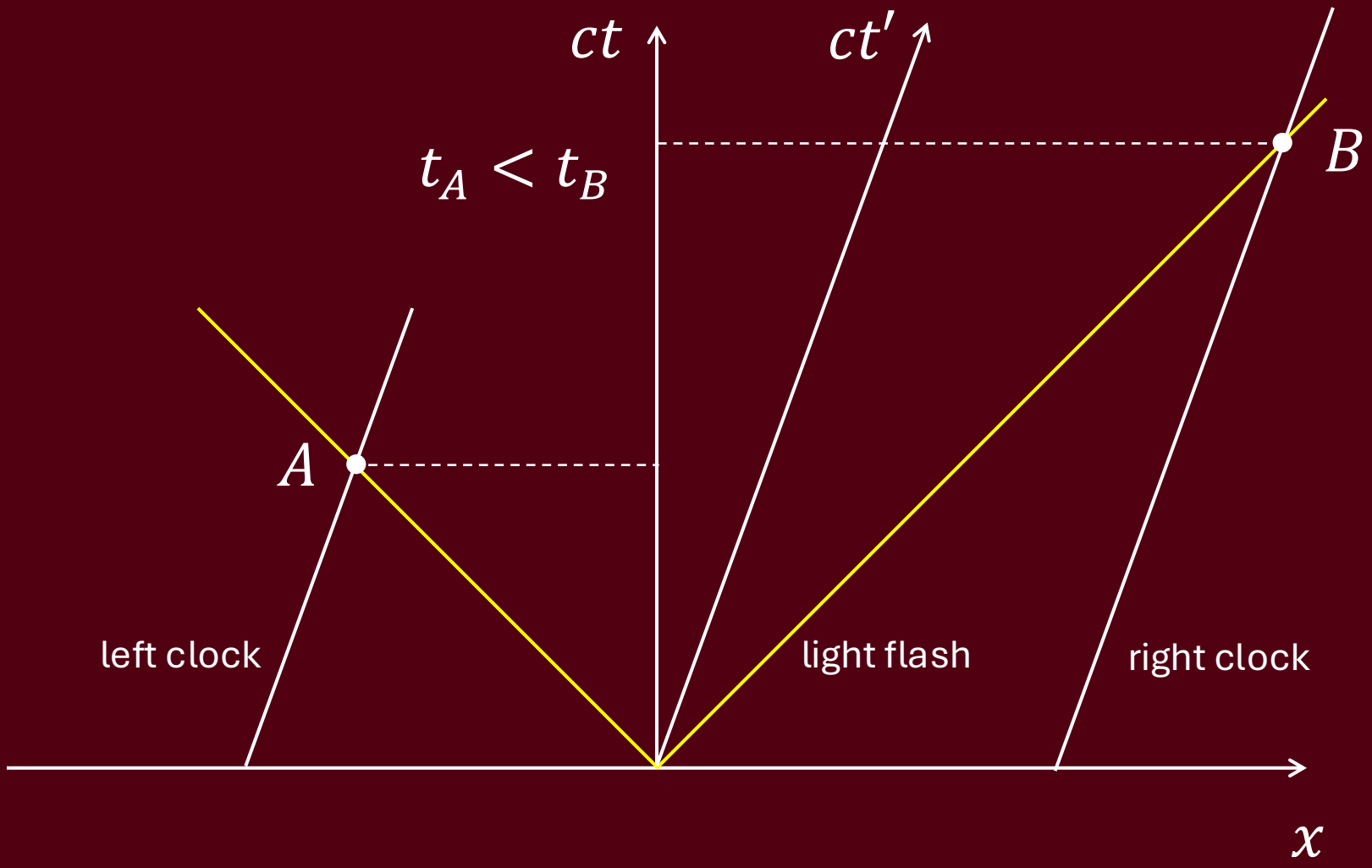
Two events that are simultaneous in the Other Frame…



Relativity of Simultaneity

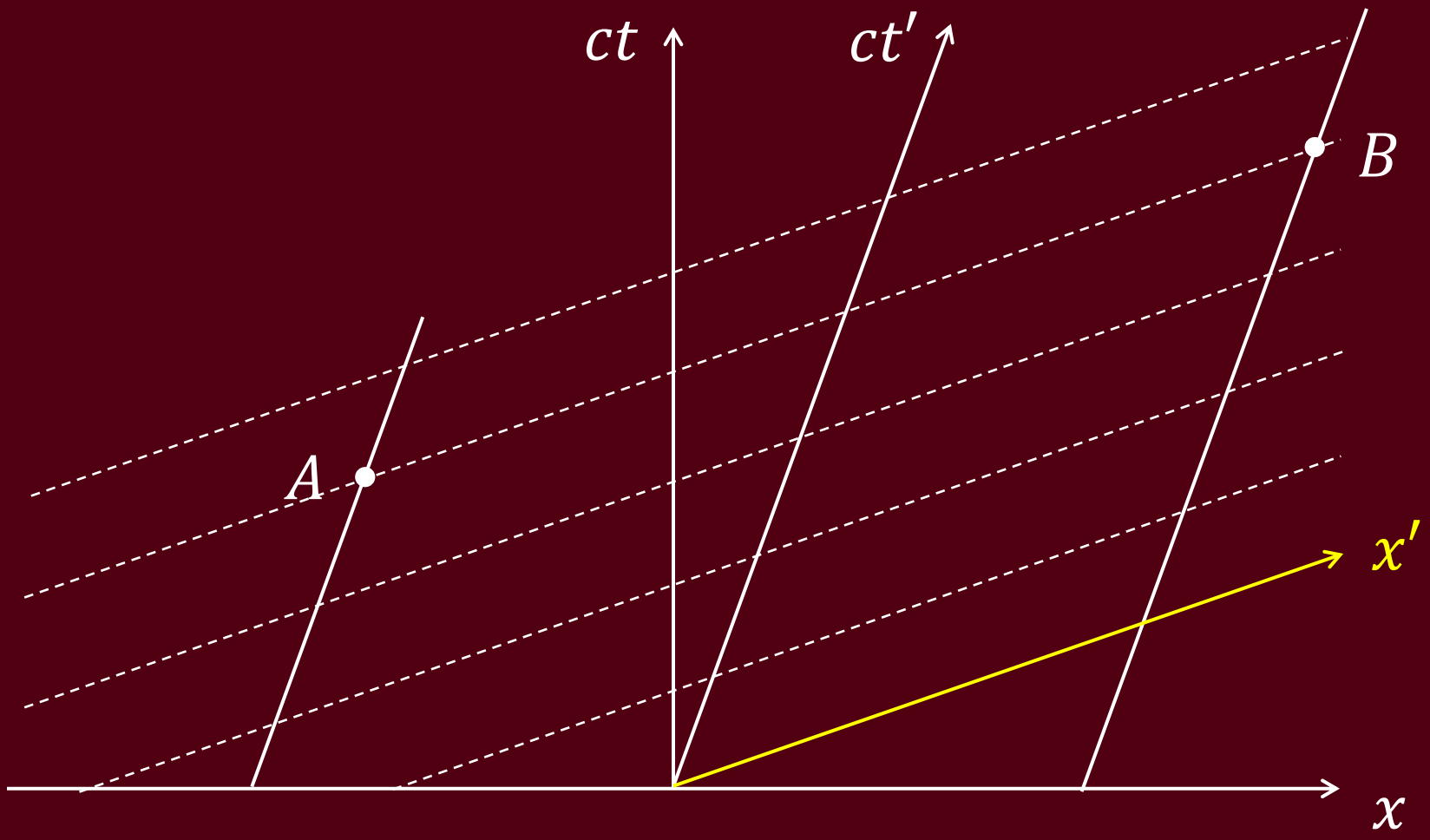
...are not simultaneous in the Home Frame





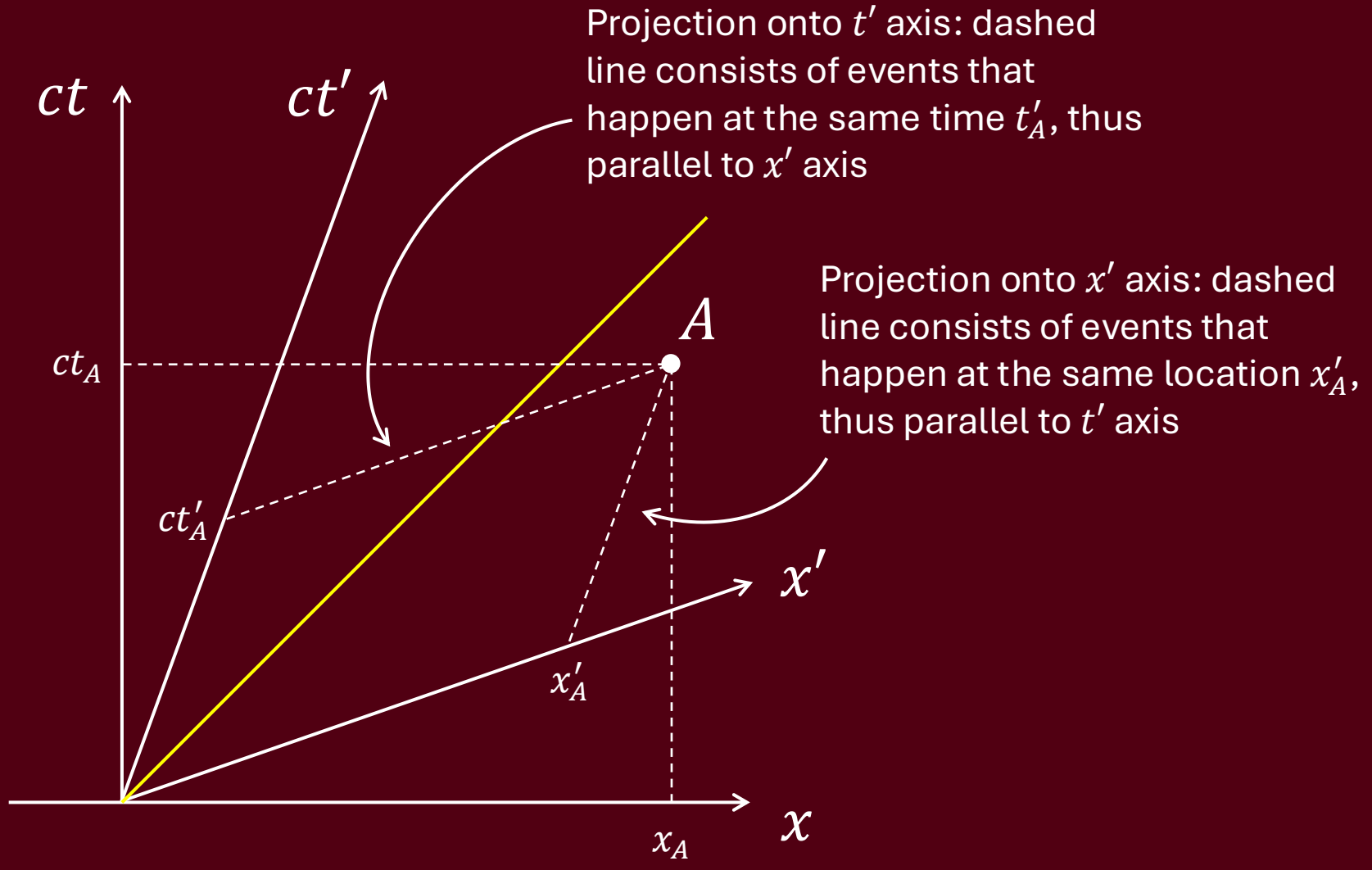
Lines of Simultaneity

The lines of simultaneity (dashed lines) connect all events that have the same time coordinate in a given frame of reference.

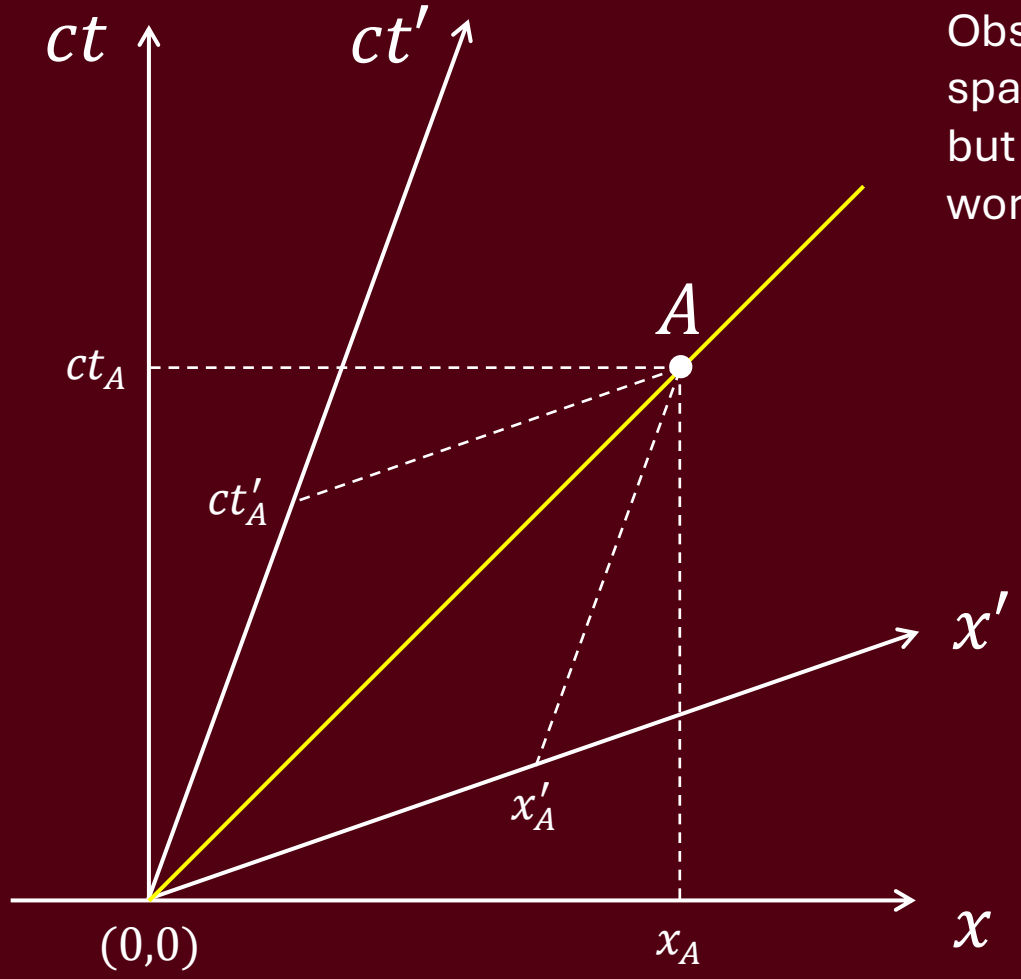


Remember, A and B are simultaneous in S'. That hasn't changed!

How to Read a Spacetime Diagram



Spacetime Diagram

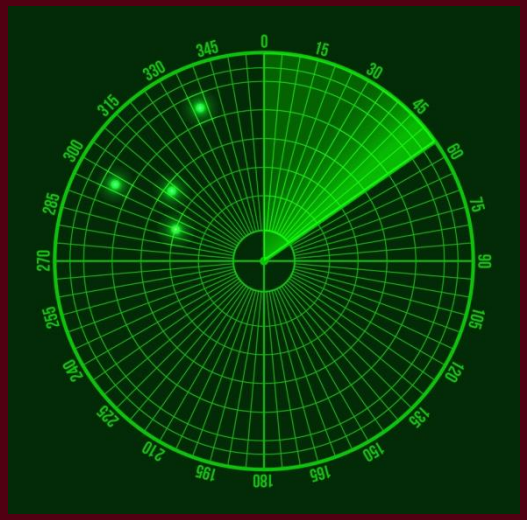


Observers disagree about the spacetime coordinates for event A, but they agree that it lies along the worldline of a light flash

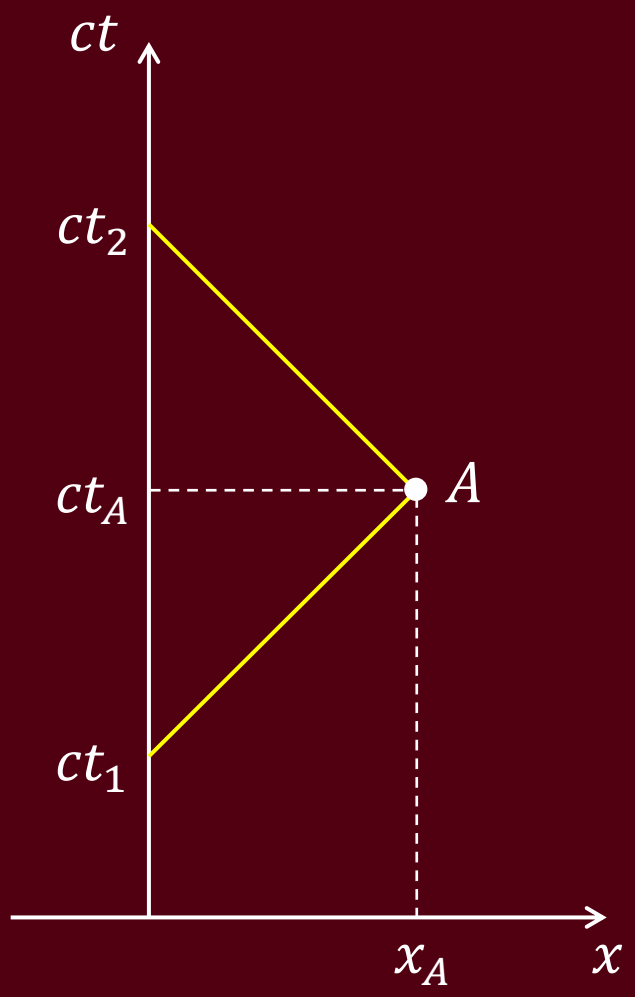
$$\frac{x_A}{t_A} = \frac{x'_A}{t'_A} = c$$

Example

Construct a spacetime diagram for the following scenario. A “master clock” sits at the origin of coordinates and periodically emits flashes of light in the $\pm x$ directions. At time t_1, the clock sends out a light flash to reflect off a target (call this event A). The reflected light travels back along the x axis to the master clock, which registers the reflected flash at time t_2. Use the diagram and basic kinematics to determine the spacetime coordinates of A.



Solutions



Light travels at constant speed c from the origin to event A, then back.

$$x_A = c(t_2 - t_A)$$

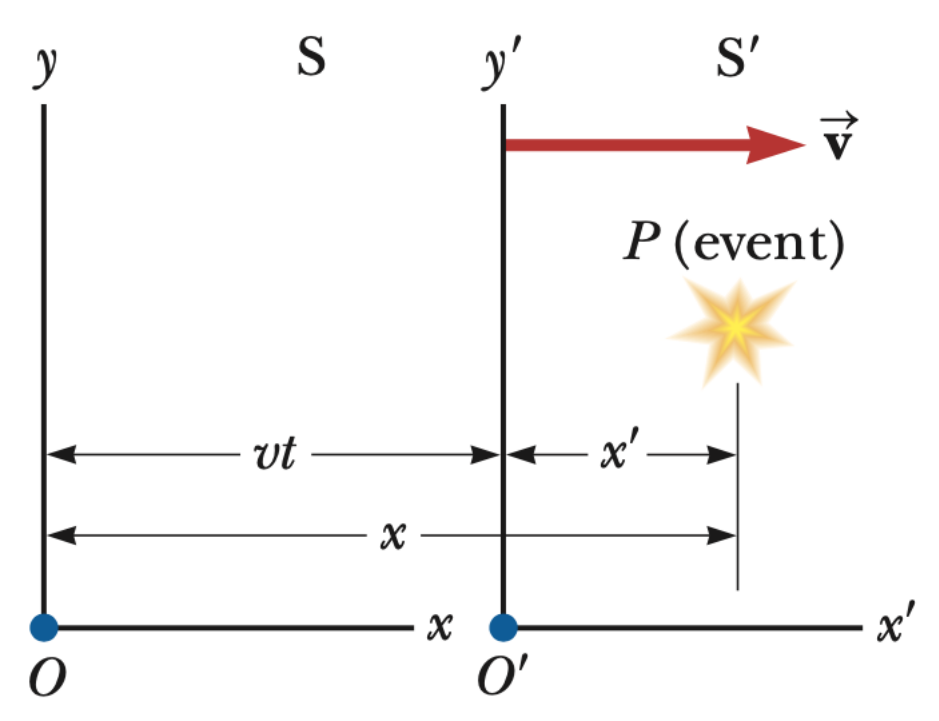
$$x_A = c(t_A - t_1)$$

$$ct_A = \frac{1}{2}(ct_2 + ct_1)$$

$$x_A = \frac{1}{2}(ct_2 - ct_1)$$

Galilean Transformation

In Newtonian mechanics, where simultaneity is absolute, it makes sense to talk about the position of an object in different frames of reference *at the same time*. This allows us to derive a set of coordinate transformations called the Galilean transformation.



$$t' = t$$

$$x' = x - vt$$

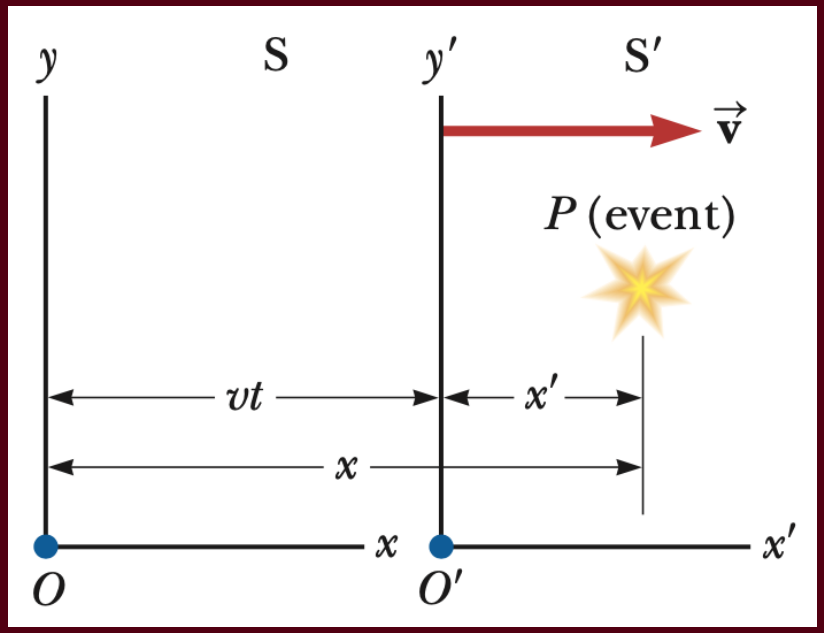
$$y' = y$$

$$z' = z$$

Good approximation when $v \ll c$

Lorentz Transformation

The Lorentz transformation is the *correct set of equations* that relate the spacetime coordinates of S and S' for all relative speeds in the range $0 < v < c$.



$$t' = \gamma(t - vx/c^2)$$

$$x' = \gamma(x - vt)$$

$$y' = y$$

$$z' = z$$

Lorentz factor: $\gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}$

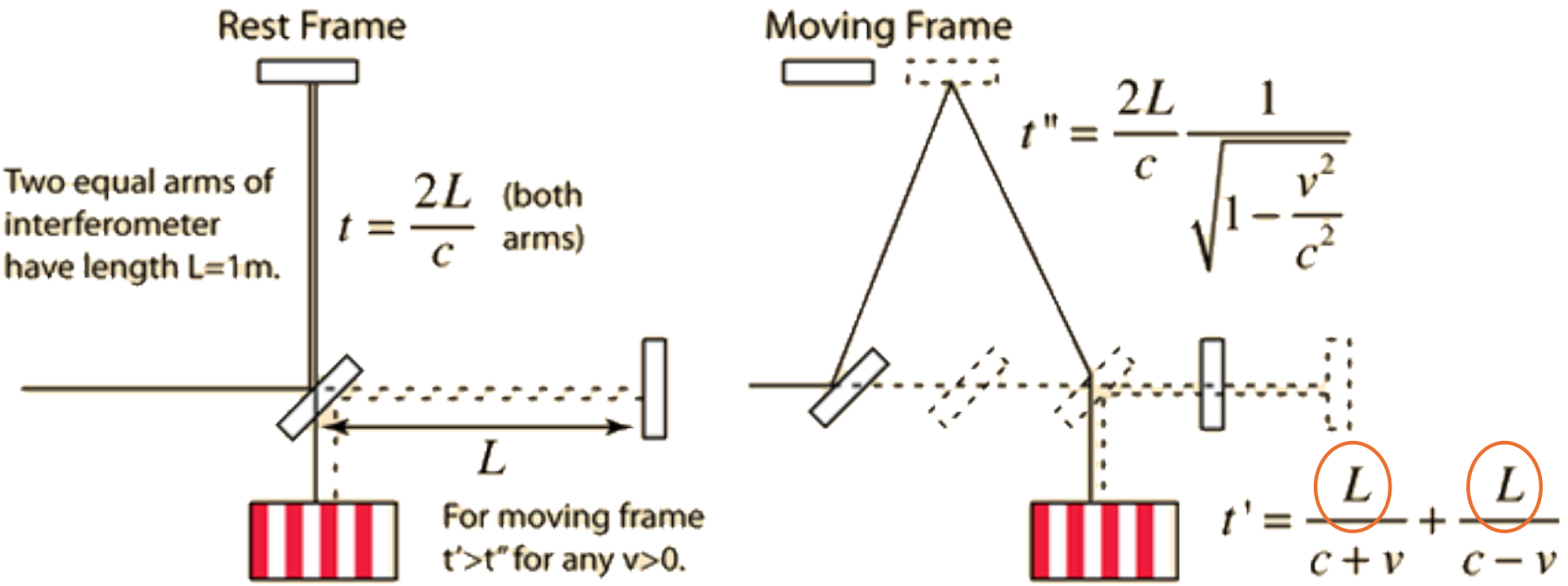
See lecture supplement on Brightspace for a derivation

Hedrik Lorentz (1853 - 1928)

- Dutch physicist who first introduced Lorentz transformation to preserve Maxwell's equations in a universe with a stationary ether.
- Knew that EM wave equation is not invariant under Galilean transformation!
- Treated time dilation and length contraction as *real but dynamical effects* caused by motion through the ether.
- Tried to explain why the Michelson-Morley experiment failed to detect motion through the ether.
- Imagine trying to measure a contracted length scale, but even the ruler you're trying to use is contracted itself due to resistive forces from the ether.







Horizontal path is shortened due to *dynamical length contraction*, according to Lorentz, so the beams always arrive in phase

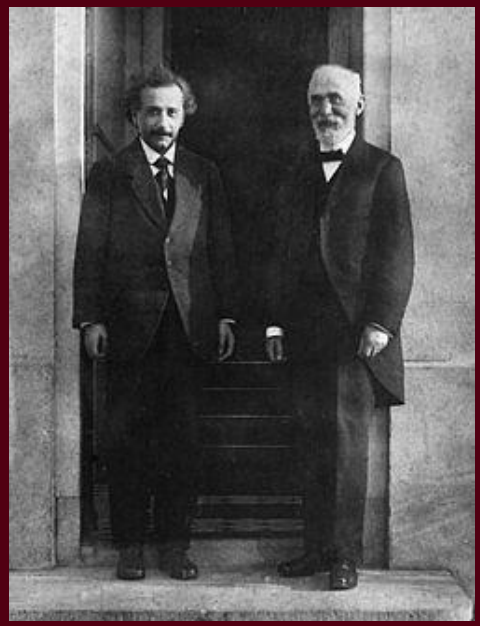
Ether vs. Special Relativity

- Lorentz essentially had the following idea: “time is absolute, the ether exists, and the laws of physics ensure we never detect it.”
  - Time dilation and length contraction arise from electromagnetic interactions due to motion relative to the ether.
- Einstein countered with: “the ether isn’t necessary; if we give up absolute time, then there are simple principles that can explain the same effects in a testable way.”
  - Time dilation and length contraction are kinematic, universal features of physical systems in relative motion.
- Special relativity is simpler, has explanatory power, unifies phenomena, and leads to new predictions.
- Sure, we give up old assumptions about absolute space and time, but we arrive at a more accurate picture of the world.

Lorentz on Einstein

"His results concerning electromagnetic and optical phenomena agree in the main with those which we have obtained in the preceding pages, the chief difference being that Einstein simply postulates what we have deduced, with some difficulty and not altogether satisfactorily, from the fundamental equations of the electromagnetic field.

By doing so, he may certainly take credit for making us see in the negative result of experiments like those of Michelson, Rayleigh and Brace, not a fortuitous compensation of opposing effects, but the manifestation of a general and fundamental principle."



Lorentz Transformation

The Lorentz transformation, as strange as it seems, can be seen as a 'natural consequence' of the principle of relativity and a few 'weak assumptions.'

- Start from the principle of relativity, assume coordinate transformations between inertial frames have sensible properties, and *don't assume a priori* that $t' = t$
- Deduce the existence of an invariant speed! Its precise value must come from experiments or some physical theory, like electromagnetism for instance.
- Deduce the Lorentz transformation!



Vladimir Ignatowski
(1875-1942)

Nehmen wir das vollständige Differential von (7) und (8), so ergibt sich

$$\left.\begin{aligned} dx' &= p\,dx + s\,dt \\ dt' &= p_1\,dx + s_1\,dt \end{aligned}\right\} \quad (9)$$

und

$$\left.\begin{aligned} dx &= p'\,dx' + s'\,dt' \\ dt &= p_1'\,dx' + s_1'\,dt' \end{aligned}\right\} \quad (10)$$

wo p, s, p', s' usw. die entsprechenden partiellen Differentialquotienten bedeuten, welche wir vorläufig noch als unbekannte Funktionen von x, t, q bezw. x', t', q' betrachten müssen.

Paar KK'' durch p'', p_1'', q_2. Dann läßt sich leicht nachweisen, daß folgende Beziehung existiert:

$$\frac{p_1}{pq} = \frac{\bar{p}_1}{pq_1} = \frac{p_1''}{p''q_2}. \quad (22)$$

$$p = \frac{1}{\sqrt{1 - q^2 n}}. \quad (24)$$

$$n = \frac{1}{c^2}.$$

Matrix Notation

It is much easier to remember the Lorentz transformation in matrix notation. Ignoring the y and z coordinates,

$$x' = \gamma(x - vt)$$

$$ct' = \gamma(ct - vx/c)$$

$$\begin{pmatrix} x' \\ ct' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -v/c \\ -v/c & 1 \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix}$$

Lorentz transformation matrix (including γ)

Matrix Multiplication

Dot product: (1st row) · (1st column) = (1st row)

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_{11}\,x + a_{12}\,y \\ a_{21}\,x + a_{22}\,y \end{pmatrix}$$

(2nd row) · (1st column) = (2nd row)

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_{11}\,x + a_{12}\,y \\ a_{21}\,x + a_{22}\,y \end{pmatrix}$$

(1st row) · (1st column) = (1st row, 1st column)

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}$$

Inverse Transformation

If S' moves right with speed v relative to S, then S moves to the *left* with speed v according to S'.

The inverse transformation from S' to S is therefore the Lorentz transformation from S to S' with $v \to -v$.

$$\begin{pmatrix} x \\ ct \end{pmatrix} = \gamma \begin{pmatrix} 1 & v/c \\ v/c & 1 \end{pmatrix} \begin{pmatrix} x' \\ ct' \end{pmatrix}$$

If you don't believe me, you're welcome to invert the matrix yourself.

Note that $\gamma(-v) = \gamma(v)$

Matrix Inverse

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{ad-bc}\begin{pmatrix} da-bc & db-bd \\ -ca+ac & -bc+ad \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$ ✅

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Lorentz Transformation & Inverse

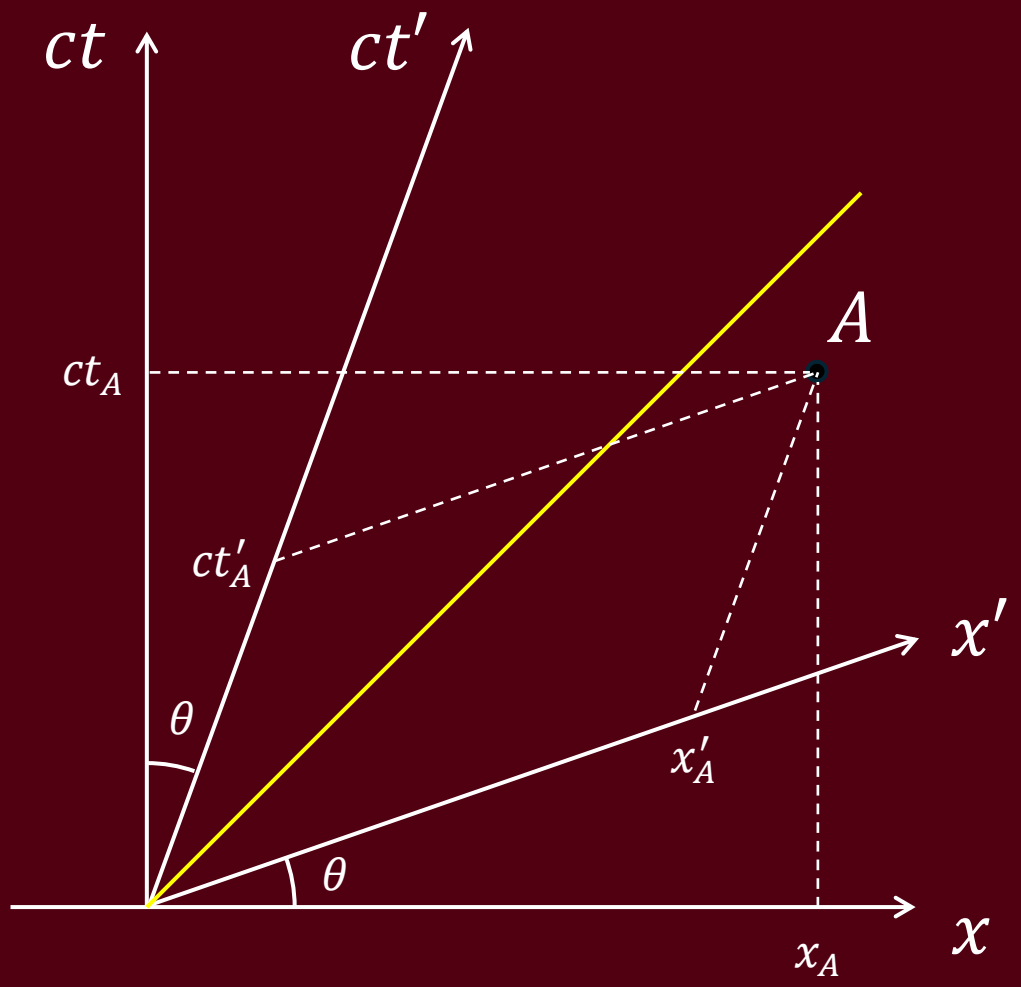
Let $\beta = v/c$ to simplify the notation. Always include a factor of c in front of the time coordinate!

Lorentz transformation

$$\begin{pmatrix} x' \\ ct' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix}$$

Inverse transformation

$$\begin{pmatrix} x \\ ct \end{pmatrix} = \gamma \begin{pmatrix} 1 & +\beta \\ +\beta & 1 \end{pmatrix} \begin{pmatrix} x' \\ ct' \end{pmatrix}$$



$$\begin{pmatrix} x_A{}' \\ ct_A{}' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} x_A \\ ct_A \end{pmatrix}$$

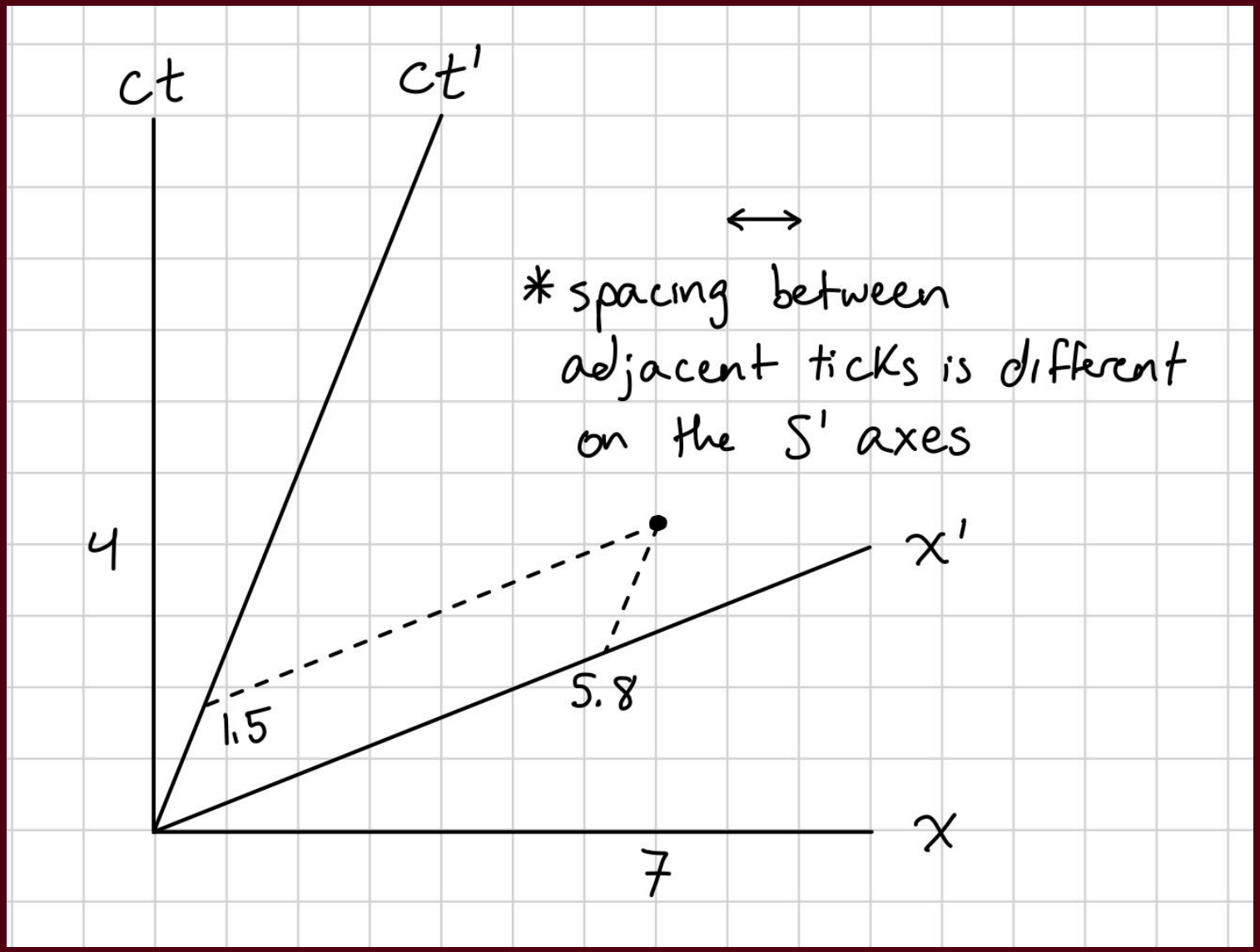
$\tan\theta = \beta$

$\tan 45° = 1$

Example

Consider an event that is observed in the Home Frame to occur at time $ct = 4.2$ m and position $x = 7.0$ m. When and where does this event occur according to observers in the Other Frame that is moving with speed $v = 2c/5$ with respect to the Home Frame?

Visualization



Example

The Other Frame measures a time interval Δt_0 between two events that occur at the origin O', i.e., $\Delta x' = 0$. What are the corresponding coordinate intervals in S?

$$\begin{pmatrix} \Delta x \\ c\Delta t \end{pmatrix} = \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} \Delta x' \\ c\Delta t' \end{pmatrix}$$

Inverse Lorentz transformation for coordinate intervals

Time Dilation

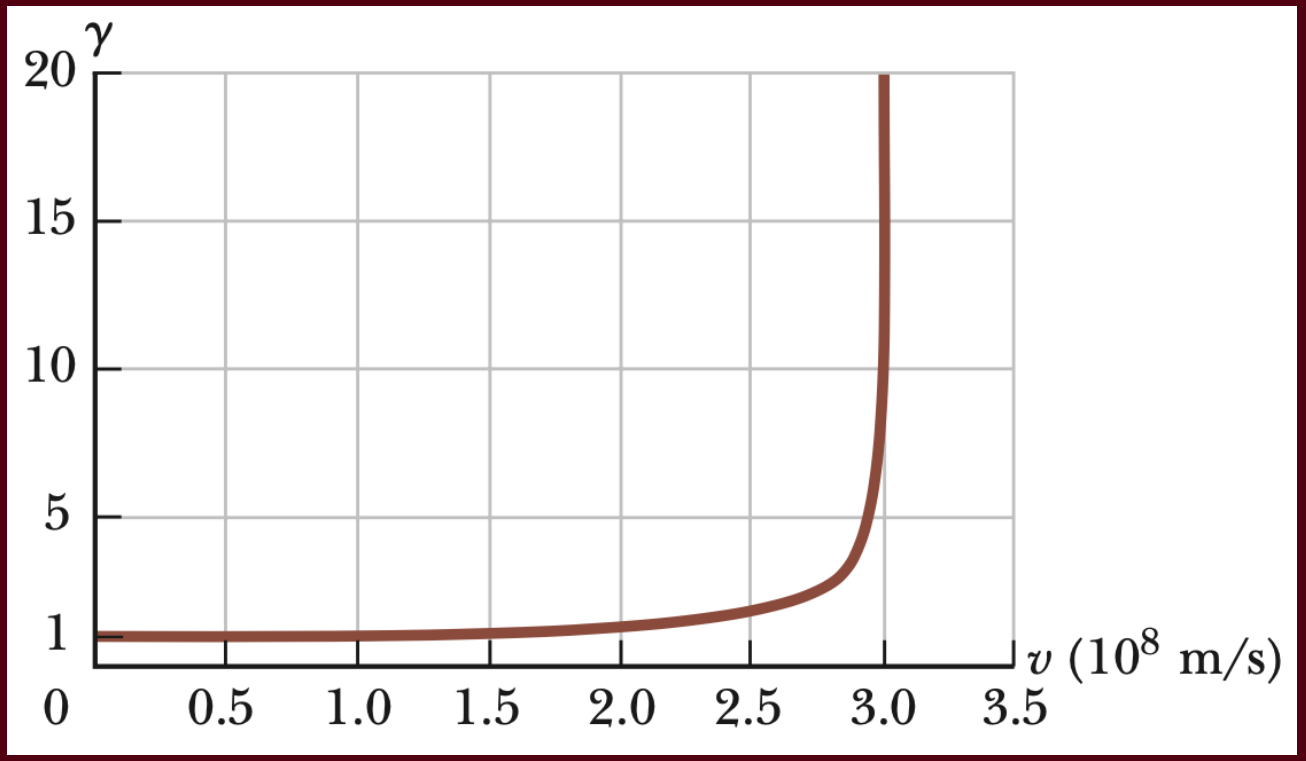
$$\Delta t = \gamma \Delta t_0 = \frac{\Delta t_0}{\sqrt{1-\beta^2}}$$

- Since $\gamma \geq 1,\ \Delta t \geq \Delta t_0$, i.e., time dilation!
- This equation also applies to situations where the speed of the clock measuring Δt_0 is constant, even if the velocity is changing (as in uniform circular motion).

Lorentz Factor

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-v^2/c^2}}$$

The Lorentz factor differs significantly from a value of 1 only for very high speeds



Useful Approximation

$$(1+x)^n = \sum_{k=0}^{\infty} \binom{n}{k} x^k = 1 + nx + \frac{n(n-1)}{2!}x^2 + \cdots$$

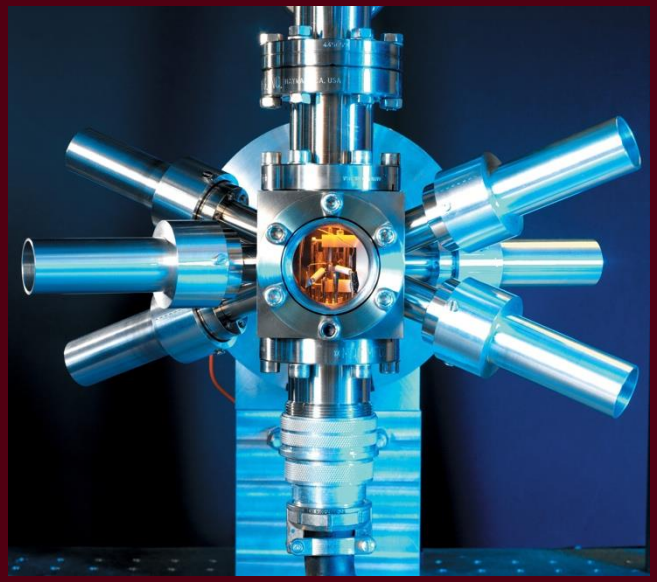
$$\gamma = (1-\beta^2)^{-1/2} = 1 + \frac{1}{2}\beta^2 + O(\beta^3)$$

For small $\beta = v/c$, with "how small" depending on the required accuracy, you can use the following approximation:

$$\gamma \approx 1 + \frac{1}{2}\beta^2$$

Example

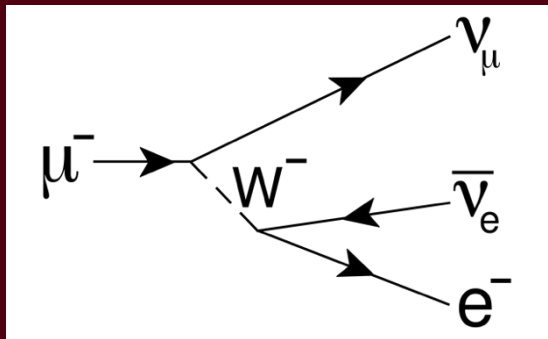
Ignoring gravitational effects and the rotation of the Earth, estimate the amount by which two atomic clocks will disagree if one clock on an airplane goes all the way around the equator at a constant speed of 300 m/s and another clock sits at rest on the Earth's surface.



Example

The half-life of an unstable particle is the amount of time it takes for one half of a large sample of identical particles to decay, on average. The half-life of a muon at rest is $1.52\ \mu s$, meaning that after $1.52\ \mu s$, a collection of N muons will decay until $N/2$ muons are remaining.

Two detectors are vertically aligned and placed 2 km apart. The detectors register only the muons traveling at $0.995c$ vertically downward in the Earth's rest frame. If N muons are detected at the top detector, what fraction are detected at the bottom one? Calculate two values, one using the Galilean transformation and another using the Lorentz transformation.



Length Contraction

Consider a stick of length L_0 at rest in the Other Frame. To measure the length L of the stick in the Home Frame, we must record the positions of each end of the stick *simultaneously* in S.

Event A has coordinates (x_A, ct) and B has coordinates (x_B, ct) such that $L = \Delta x = x_B - x_A$

$$\begin{pmatrix} L_0 \\ c\Delta t' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} L \\ 0 \end{pmatrix}$$

$$L = \frac{L_0}{\gamma} = L_0\sqrt{1-\beta^2}$$

Example

Why don't we observe time dilation or length contraction in our everyday lives?

The simple answer is that we typically transform quantities between reference frames moving with relative speeds much slower than light.

Show that the Lorentz transformation reduces to the Galilean transformation in the limit $v \ll c$.

Spacetime Interval

Consider any two events in spacetime. We define the spacetime interval such that

$$\Delta s^2 = (c\Delta t)^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$$

The spacetime interval is invariant under Lorentz transformations, in a way that is analogous to the invariance of length under rotations.

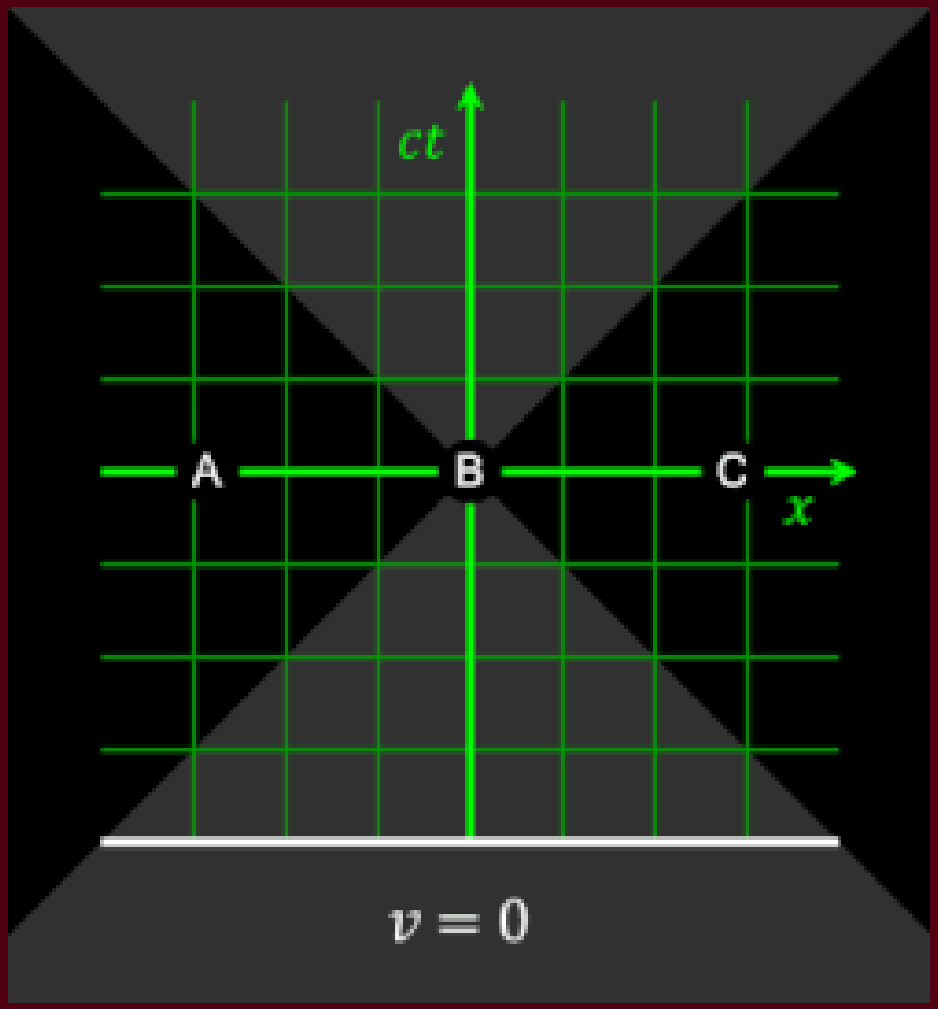
Invariance means $\Delta s^2 = \Delta s'^2$ for any two inertial frames in relative motion. The equation for Δs^2 is often called a “metric equation.”

Spacetime Interval

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$$

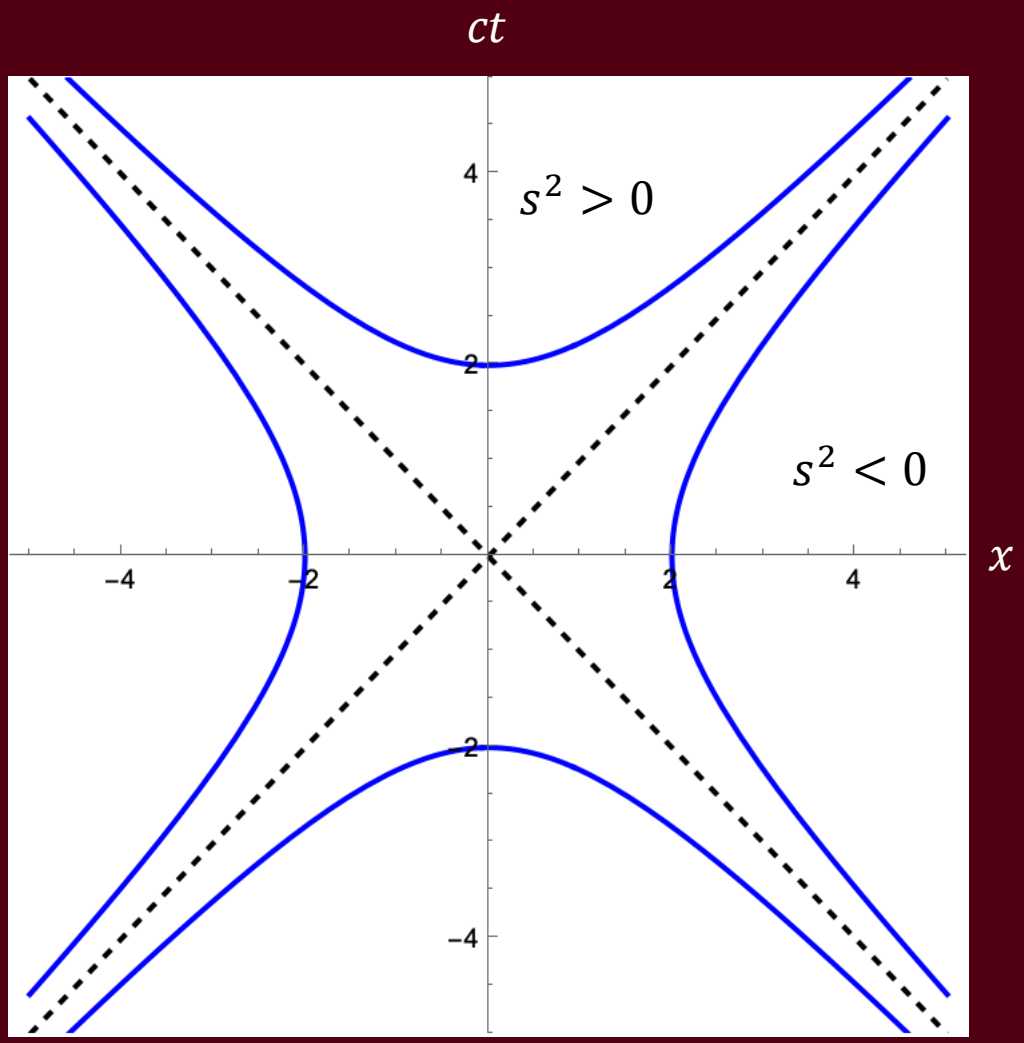
The invariant spacetime interval is a generalization of the concept of length applied to 4-dimensional spacetime. We classify the separation of events depending on the sign of Δs^2.

- Timelike: when $\Delta s^2 > 0$, there exists a frame of reference where $\Delta\tau = \Delta s/c$ is a proper time interval $(\Delta r = 0)$. Observers in relative motion agree on the ordering of timelike separated events
- Spacelike: when $\Delta s^2 < 0$, there exists a frame of reference where $\Delta\ell = \sqrt{-\Delta s^2}$ is a proper length $(\Delta t = 0)$. Observers in relative motion may disagree on the ordering of spacelike separated events
- Lightlike: $\Delta s^2 = 0$, a light flash could travel between the events


ct
A
B
C
x
v = 0

Lines of constant s^2 are hyperbolas

$$c^2t^2 - x^2 = s^2$$



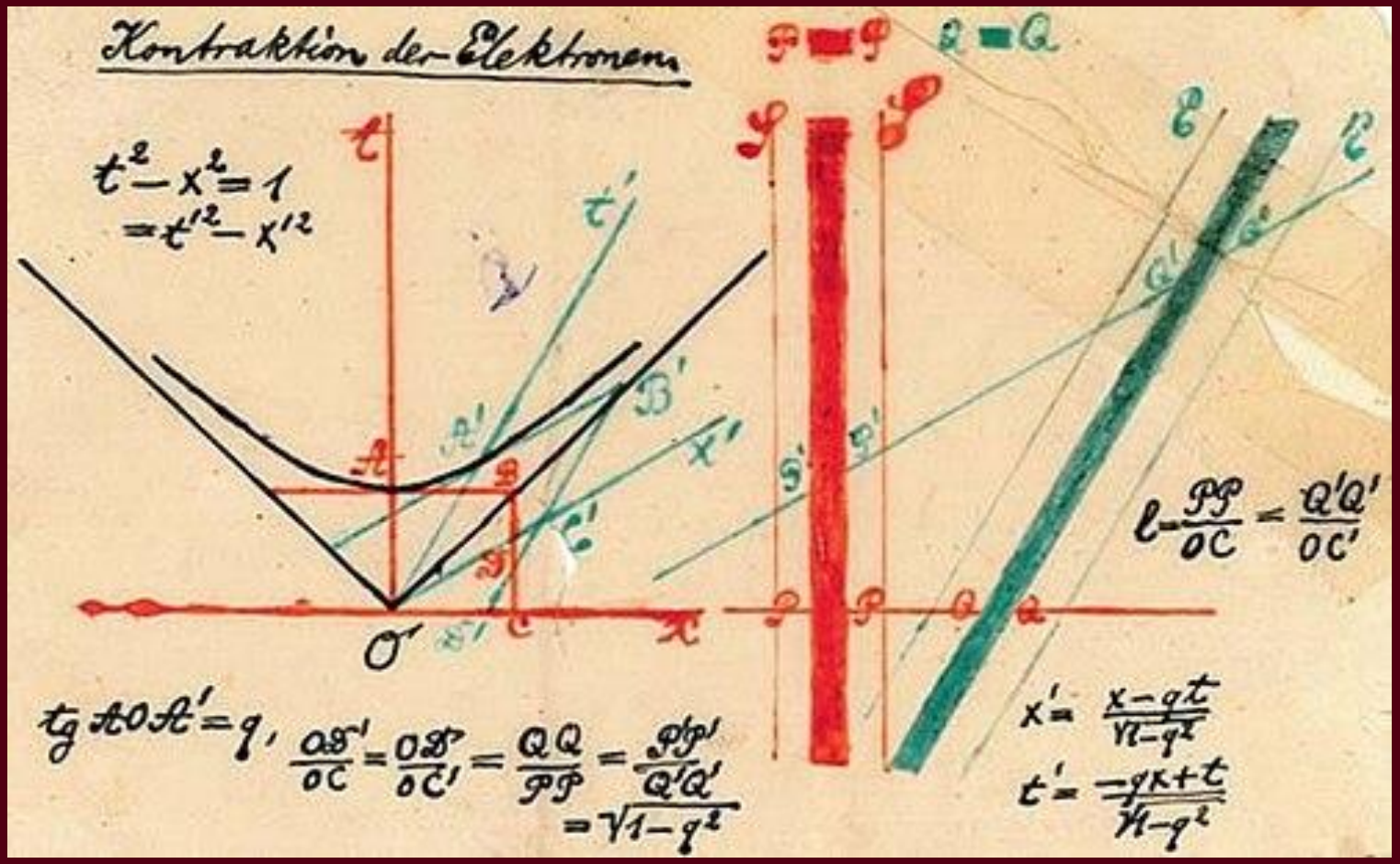
Herman Minkowski (1864 – 1909)

- Provided a geometric interpretation of Einstein's special relativity based on 4-dimensional spacetime, later called Minkowski spacetime.
- Interpreted Lorentz transformations as 'rotations' in 4-dimensional spacetime that preserve the spacetime interval between events.
- Einstein initially thought Minkowski's work was "superfluous," but later found that a geometric interpretation was essential to his formulation of general relativity in 1915.



“The views of space and time which I wish to lay before you have sprung from the soil of experimental physics, and therein lies their strength. They are radical. *Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality.*”

- Minkowski


Kontraktion der Elektronen
$t^2 - x^2 = 1$
$= t'^2 - x'^2$
$P = p \quad Q = Q$
$l = \frac{PP}{OC} = \frac{Q'Q'}{OC'}$
$tg\, AOA' = q, \quad \frac{OS'}{OC} = \frac{OS}{OC'} = \frac{QQ}{PP} = \frac{PP'}{Q'Q'} = \sqrt{1-q^2}$
$x' = \frac{x - qt}{\sqrt{1-q^2}}$
$t' = \frac{-qx + t}{\sqrt{1-q^2}}$

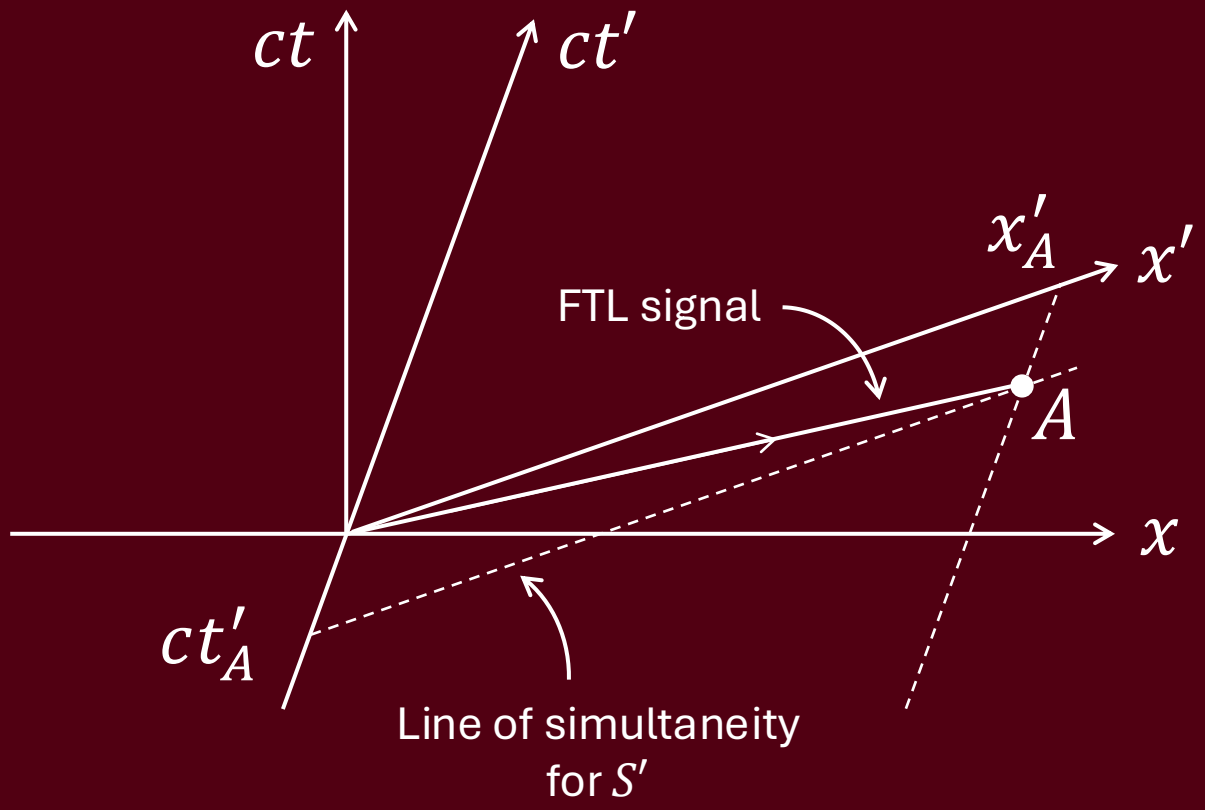
Cosmic Speed Limit

"Nothing can go faster than light."

There are different ways to understand this result, but for one, it must be true to preserve causality, which is the idea that a causal event must precede its effect. If faster-than-light travel were possible, effects could precede their causes, and that's absurd!

The speed of light is the "cosmic speed limit" in the sense that no physical object capable of exerting causal influence can travel faster than c.

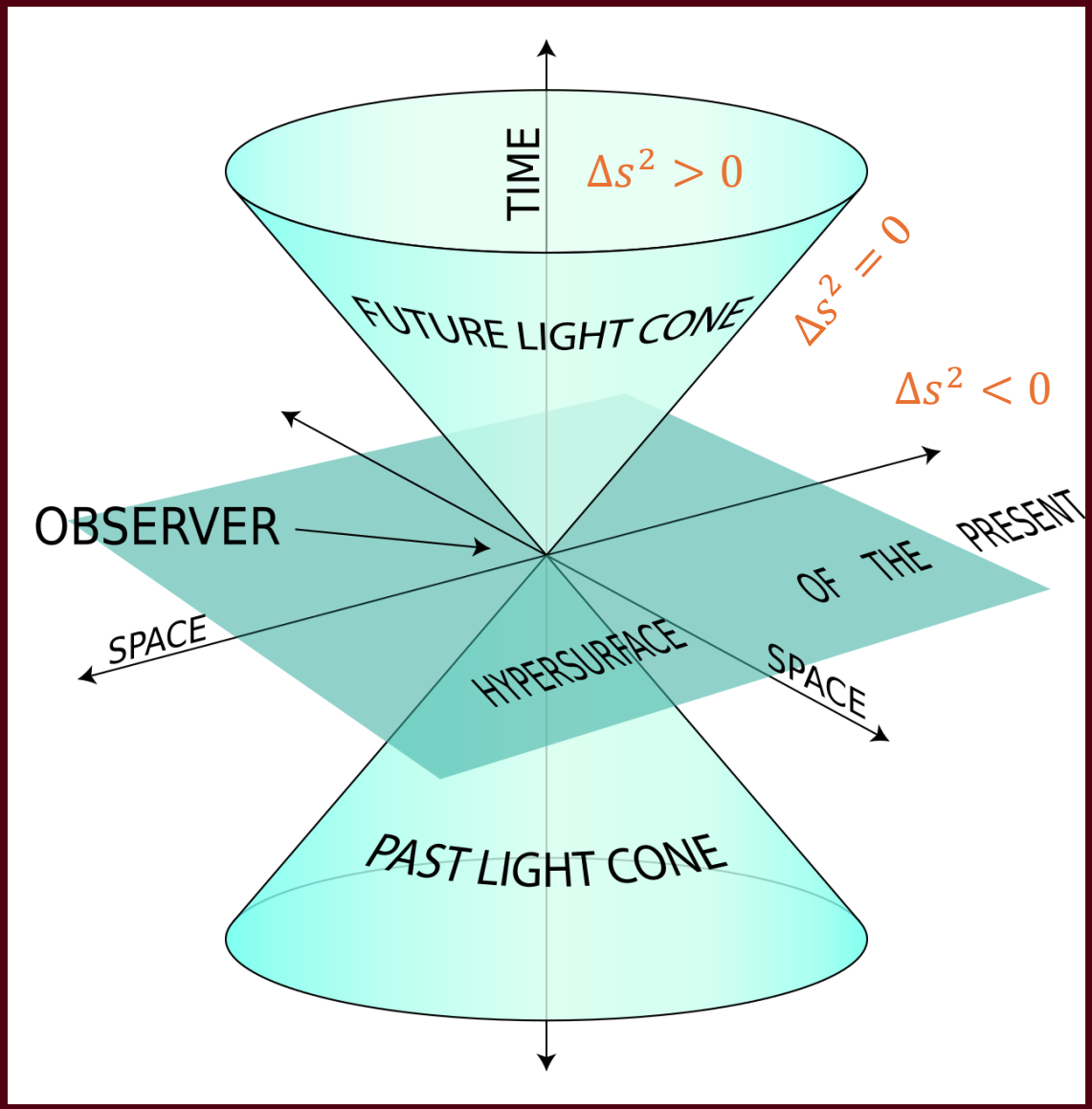
Proof by Contradiction



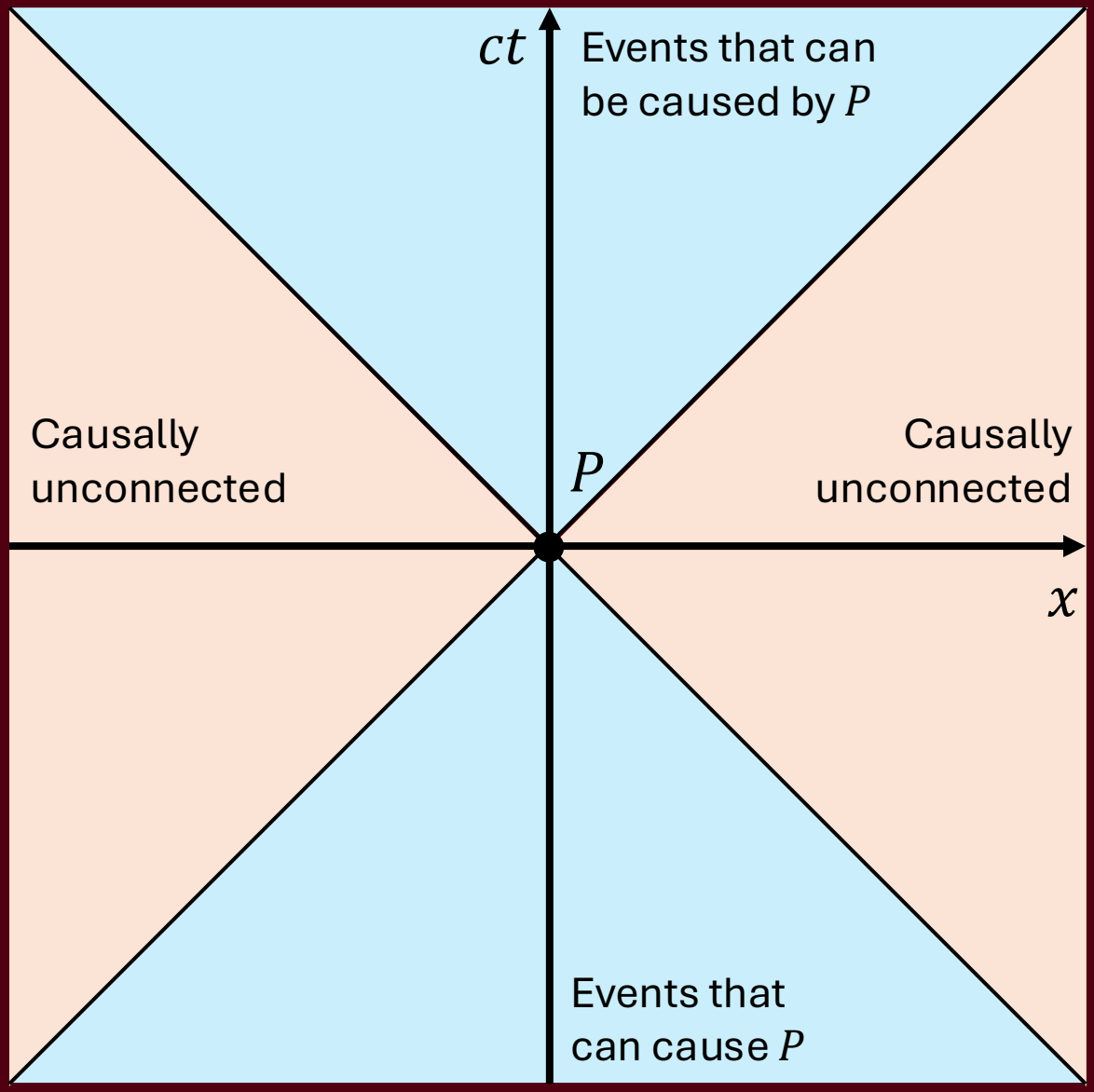
In the Home Frame, the cause is at the origin,
and its hypothetical effect is event A.


186,000 miles per second. It's not just a good idea...
It's the law!

Light Cone


TIME
$\Delta s^2 > 0$
FUTURE LIGHT CONE
$\Delta s^2 = 0$
$\Delta s^2 < 0$
OBSERVER
SPACE
HYPERSURFACE
OF THE PRESENT
SPACE
PAST LIGHT CONE

Light Cone


ct
Events that can be caused by P
Causally unconnected
P
Causally unconnected
x
Events that can cause P

Conceptual Question

A meteor strikes the moon (event A), causing a large and vivid explosion. Exactly 0.47 s later (as measured in an inertial reference frame attached to the earth) the radio telescope receiving signals from the moon goes on the fritz. Could these events be causally related? (Hint: The distance between the earth and the moon is roughly 384,000 km.)



